LIG: Layer-wise Integrated Gradients for Within-Layer Flow Analysis in Transformers

MON-19

Poster (PDF) | My page

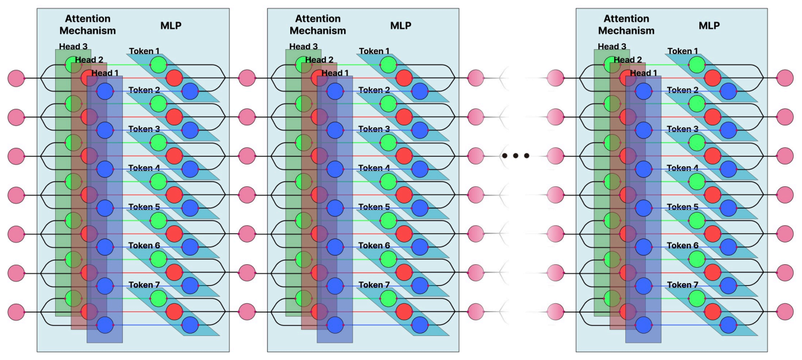
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Attributing ATT and MLP separately and composing them matches whole-layer IG best when their baselines follow the whole-layer baseline.

LIG (Layer-wise Integrated Gradients) applies Integrated Gradients (IG) at the Multi-Head Attention (ATT) and multilayer perceptron (MLP) boundaries and composes them. Best on BERT-base, 1,700 Penn Treebank (PTB) sentences: whole-layer ITB → ITB for ATT, ATTITBa=0 for MLP; whole-layer Zero → Zero for both.

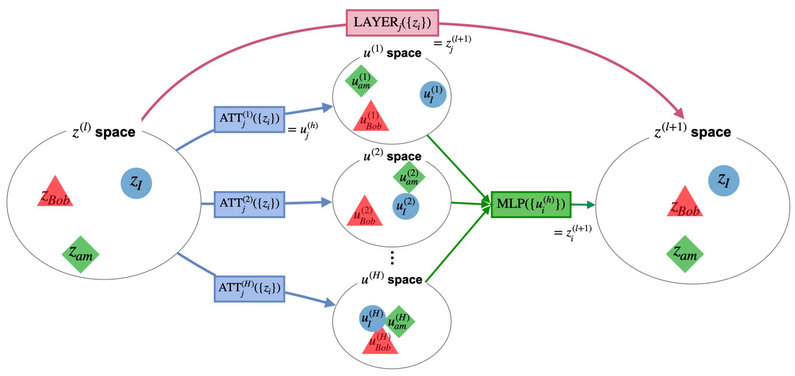
Project page
https://eightsuzuki.github.io/layer-wise-integrated-gradients/

1 What flows inside a layer?



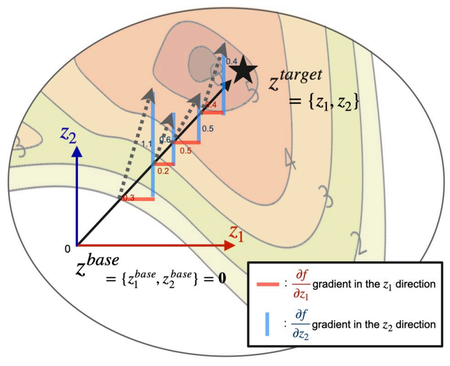
Each layer as a **dynamic graph**: tokens z_i (pink) and per-head attention outputs $u_i^{(h)}$ are nodes; information crosses the ATT and MLP boundaries.

We attribute each boundary with IG (bottom path) and ask whether the composition matches IG on the whole layer (top path).



Method	Limitation / LIG's gain
PRIOR Attention map [7]	ignores values and MLP
PRIOR Rollout [5]	ignores values and MLP
PRIOR Norm-based [6]	no ATT / MLP split
PRIOR AttnLRP [4]	a rule per operation
OURS **LIG**	**one rule (IG); splits and composes**

2 IG, and where the path starts



PRIOR IG [1] *line-integrates* the gradient along a path from a **baseline** $z(0)$ to the input z. Any path keeps completeness, so path and start are ours to choose; we use the straight path $z(a) = z(0) + a\,(z - z(0))$:

$$\mathrm{IG}_d = (z_d - z_d(0)) \int_0^1 \frac{\partial f(z(a))}{\partial z_d}\, da$$

f: scalar output (contours); d: input dimension.

Completeness: $\sum_d \mathrm{IG}_d = f(z) - f(z(0))$. The baseline sets **what each contribution is measured against** (§5).

3 LIG: set-to-set IG at each boundary OURS

At each boundary (cells below), apply IG to the input x from its **baseline** $x(0)$ with $f = \|y_j(a) - y_j(0)\|_2$, summing over each input element r (token or head); scores are signed:

$$\mathrm{IG}_{r,j} = \sum_{d \in r} \mathrm{IG}_d, \qquad \widetilde{\mathrm{IG}}_{r,j} = \frac{\mathrm{IG}_{r,j}}{\|y_j(1) - y_j(0)\|_2} \quad \left(\sum_r \widetilde{\mathrm{IG}}_{r,j} = 1\right)$$

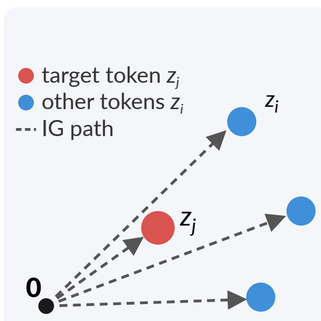
ATT
$z_i \to u_j^{(h)}$: $\mathrm{IG}^{\mathrm{ATT}}_{i,j,h}$

MLP (W_o, LayerNorm, FFN; residual z fixed)
$u_j^{(h)} \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{MLP}}_{h,j}$

Whole layer l
$z_i \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{LAYER}}_{i,j}$

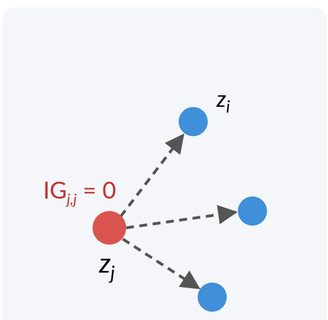
Baselines: how the **input** moves along the IG path, $a = 0 \to 1$ (input-space sketch).

Zero



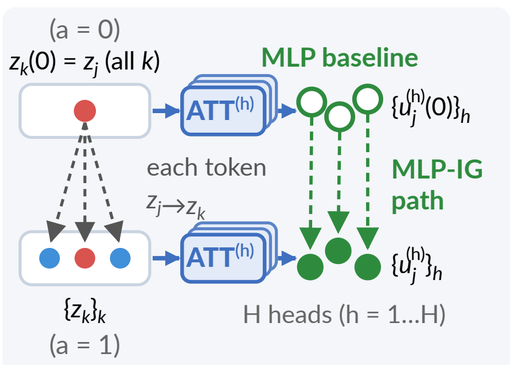
Every path starts at the zero vector $\mathbf{0}$.

Input Token Base (ITB)



All tokens start at the target z_j; z_j itself does not move, so $\mathrm{IG}_{j,j} = 0$.

MLP: ATTITBa=0



At $a = 0$ all tokens are z_j, so each head outputs z_j's value $u_j^{(h)}(0)$: the MLP baseline.

Why ITB? ATT averages token values, so we measure each token *relative to the target* z_j. Its zero self term [8] comes from the reference, not the layer (self-attention $\alpha_{j,j} > 0$); zeroRatio (from Zero-baseline scores) or mapRatio (from $\alpha_{j,j}$) estimates it, keeping the total.

Why ATTITBa=0? MLP inherits the ATT output at the ITB start ($a = 0$), keeping both modules on one reference.

4 Composing ATT and MLP OURS

Chain the two boundaries as in LRP [2, 3] and compare the composed token-level allocation with whole-layer IG:

$$\underbrace{\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j}}_{\text{whole layer}} \overset{?}{\approx} \underbrace{\sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}}_{\text{composed}}$$

Not an IG chain rule: the composition omits the residual route, and separately integrated paths need not agree. We measure this *compositional gap*, not faithfulness.

5 Result: does the composition match the layer?

Setup. BERT-base-uncased; PTB dev, 1,700 sentences [9]; $m = 32$ steps. Per sentence, layer l and output token j, the L_2 distance $d_j^{(l)}$ between the two sides of §4 (lower = better).

Mean over (sentence, layer, token); top 3 per block, **bold** = best; compare within blocks.

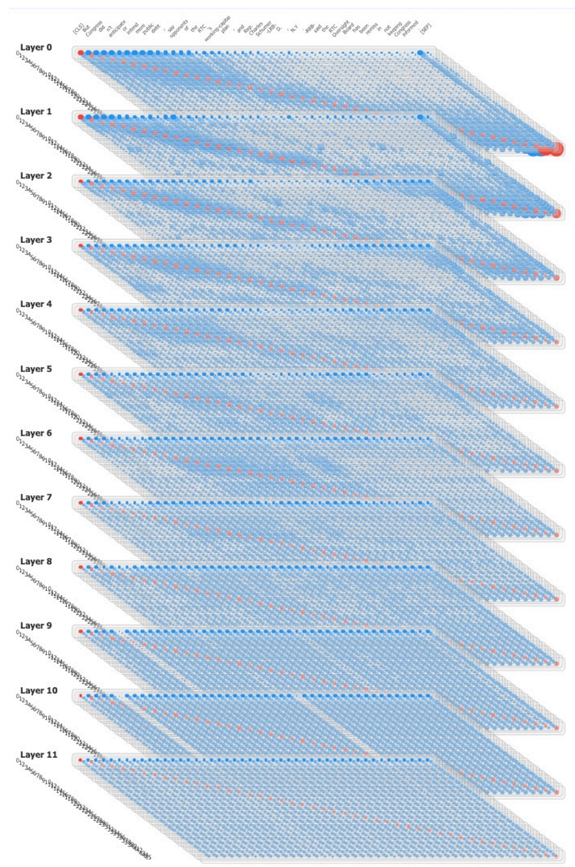
Whole-layer baseline (reference)	Composed: ATT	MLP	$L_2 \downarrow$
ITB Input Token Base: all tokens start at z_j	**ITB**	**ATTITBa=0**	**0.305**
	ITB-zeroRatio	ATTITBa=0	0.312
	ITB-mapRatio	ATTITBa=0	0.318
Zero zero vector $\mathbf{0}$	**Zero**	**Zero**	**0.736**
	ITB-zeroRatio	Zero	0.744
	Zero	ATTITBa=0	0.749

> The best MLP baseline follows the reference: ATTITBa=0 under ITB (for every tested ATT baseline); Zero under Zero.

6 Use case: reading the flow layer by layer

Whole-layer IG, Zero baseline, layers 0–11, one PTB sentence (qualitative). Columns: input tokens; rows: output tokens; circle size = contribution; red = self, blue = other tokens.

Layer 0: mass on each token and its neighbors; **layers 9–11:** spread across the sentence.



7 Scope and next steps

- **Limits:** BERT-base / PTB only; straight paths may leave the data distribution; ITB completion uniqueness is open; §6 is qualitative; L_2 discards direction. Needing only module boundaries and gradients, LIG can extend to other architectures (untested).
- **Next:** decoder-only models; comparison with EAP / EAP-IG [10, 11]; intervention-based checks.

[1] Sundararajan+ ICML 2017 · [2] Bach+ PLOS ONE 2015 · [3] Montavon+ LRP overview 2019 · [4] Achtibat+ (AttnLRP) ICML 2024 · [5] Abnar & Zuidema ACL 2020 · [6] Kobayashi+ EMNLP 2020 · [7] Jain & Wallace NAACL 2019 · [8] Sturmfels+ Distill 2020 · [9] Marcus+ Treebank-3, LDC 1999 · [10] Syed+ BlackboxNLP 2024 · [11] Hanna+ COLM 2024. Supported by JSPS KAKENHI JP26K02989, JP26K23861, JP25K24611.